Hao Zhou, Ph.D.

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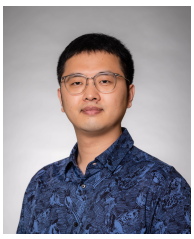
Traffic flow theory; complex network science; self-driving design and experiment; deep reinforcement learning



Work

Aug 2023 – present	**Assistant Professor**, University of South Florida
Sep 2022 – July 2023	**Postdoc research fellow** , Georgia Tech
Aug 2018 – Aug 2022	**Graduate research assistant** , Georgia Tech

Education

2018 – 2022 **Ph.D., Georgia Institute of Technology**, Civil Engineering.
Dissertation: *Longitudinal Control of Self-driving Cars Based on Traffic Flow Considerations: Theory, Design and Experiments (Advisor: Jorge Laval)*

2015 – 2018 **M.S., Southeast University**, Transportation Engineering.
Thesis: *Operational Characteristics of Exiting Trucks at Freeway Diverges Based on Drone Video Processing (Advisor: Jianchuan Cheng)*

2011 – 2015 **B.S., Southeast University**, Road and Railway Engineering, School of Transportation.
Thesis: *Bernoulli Effect of Strong Winds and Its Impact on Bus Driving Safety* .

Publications

Published Journal Articles

1. Zhou, A., Peeta, S., **Zhou**, **H.**, Laval, J., Wang, Z., & Cook, A. (2024). Implications of stop-and-go traffic on training learning-based car-following control. *Transportation Research Part C: Emerging Technologies*, 104578.
2. Wang, S., Ma, Y., Easa, S. M., **Zhou**, **H.**, Lai, Y., & Chen, W. (2023). Sight distance of automated vehicles considering highway vertical alignments and its implications for speed limits. *IEEE Intelligent Transportation Systems Magazine*.
3. Laval, J., & **Zhou**, **H.** (2022). Congested urban networks tend to be insensitive to signal settings: Implications for learning-based control. *IEEE Transactions on Intelligent Transportation Systems*.
4. Li, T., Chen, D., **Zhou**, **H.**, Xie, Y., & Laval, J. (2022). Fundamental diagrams of commercial adaptive cruise control: Worldwide experimental evidence. *Transportation Research Part C: Emerging Technologies*, *134*, 103458.
5. **Zhou**, **H.**, Toth, C., Guensler, R., & Laval, J. (2022). Hybrid modeling of lane changes near freeway diverges. *Transportation Research Part B: Methodological*, *165*.
6. **Zhou**, **H.**, Zhou, A., Laval, J., Liu, Y., & Peeta, S. (2022). Incorporating driver relaxation into factory adaptive cruise control to reduce lane-change disruptions. *Transportation Research Record*, 03611981221085517.
7. **Zhou**, **H.**, Zhou, A., Li, T., Chen, D., Peeta, S., & Laval, J. (2022a). Congestion-mitigating mpc design for adaptive cruise control based on newell's car following model: History outperforms prediction. *Transportation Research Part C: Emerging Technologies*, *142*.

8. **Zhou, H.**, Zhou, A., Li, T., Chen, D., Peeta, S., & Laval, J. (2022b). Significance of low-level control to string stability under adaptive cruise control: Algorithms, theory and experiments. *Transportation Research Part C: Emerging Technologies*, *140*, 103697.

9. Li, T., Chen, D., **Zhou, H.**, Laval, J., & Xie, Y. (2021). Car-following behavior characteristics of adaptive cruise control vehicles based on empirical experiments. *Transportation research part B: methodological*, *147*, 67–91.

10. **Zhou, H.**, Laval, J., Zhou, A., Wang, Y., Wu, W., Qing, Z., & Peeta, S. (2021). Review of learning-based longitudinal motion planning for autonomous vehicles: Research gaps between self-driving and traffic congestion. *Transportation Research Record*, 03611981211035764.

11. Chen, S., Tang, Z., **Zhou, H.**, & Cheng, J. (2019). Extracting topographic data from online sources to generate a digital elevation model for highway preliminary geometric design. *Journal of Transportation Engineering, Part A: Systems*, *145*(4), 04019003.

12. **Zhou, H.**, Wang, L., Cheng, J., & Ma, Y. (2017). Numerical study of bernoulli effect on bus controllability and driving safety. *Transportation Research Record*, *2637*(1), 74–82.

Refereed Conference Proceedings

1. Zhou, A., **Zhou, H.**, Peeta, S., & Laval, J. (2023). Impact of training data set on learning-based, car-following control: Insights from parameter estimation. In *Proceedings of the transportation research board 102th annual meeting*. Transportation Research Board Washington, DC.

2. **Zhou, H.**, Zhou, A., Peeta, S., & Laval, J. (2023). Empirical study of acceleration and deceleration constraints in commercial adaptive cruise control and their impact on traffic flow. In *Proceedings of the transportation research board 102th annual meeting*. Transportation Research Board Washington, DC.

3. Li, T., Chen, D., **Zhou, H.**, & Laval, J. (2022a). Analysis of adaptive cruise control vehicles' car-following features based on empirical experiments. In *Proceedings of the transportation research board 101th annual meeting*. Transportation Research Board Washington, DC.

4. Li, T., Chen, D., **Zhou, H.**, & Laval, J. (2022b). Hysteresis in adaptive cruise control vehicles: A study based on empirical experiments. In *Proceedings of the transportation research board 101th annual meeting*. Transportation Research Board Washington, DC.

5. Zhou, A., **Zhou, H.**, Peeta, S., & Laval, J. (2022). String instability mitigation in adaptive cruise control systems using trajectory shaper and linear acceleration/deceleration relaxation. In *Proceedings of the transportation research board 101th annual meeting*. Transportation Research Board Washington, DC.

6. **Zhou, H.**, Zhou, A., Ding, Z., Laval, J., & Peeta, S. (2022). Empirical study on the acceleration/deceleration constraints under commercial adaptive cruise control. In *2022 ieee 25th international conference on intelligent transportation systems (itsc)* (pp. 3069–3076). IEEE.

7. **Zhou, H.**, Zhou, A., Li, T., Chen, D., Peeta, S., & Laval, J. (2022c). Significance of low-level controller for string stability under adaptive cruise control. In *Proceedings of the transportation research board 101th annual meeting*. Transportation Research Board Washington, DC.

8. Li, T., Chen, D., **Zhou, H.**, & Laval, J. (2021). On the equilibrium and non-equilibrium features of adaptive cruise control vehicles: A study based on empirical experiments. In *Proceedings of the transportation research board 100th annual meeting*. Transportation Research Board Washington, DC.

9. **Zhou, H.**, & Laval, J. (2021). Avoiding gridlock on large congested networks: A multi-agent deep reinforcement learning approach with spillback knowledge. In *Proceedings of the transportation research board 100th annual meeting*. Transportation Research Board Washington, DC.

10. **Zhou, H.**, Zhou, A., Li, T., Chen, D., Peeta, S., & Laval, J. (2021). Low-level controllers play a decisive role in the string stability of adaptive cruise control. In *2021 ieee international intelligent transportation systems conference (itsc)* (pp. 891–898). doi:10.1109/ITSC48978.2021.9564679

11. Laval, J., & **Zhou, H.** (2020). Large-scale traffic signal control using machine learning: Some traffic flow considerations. In *Proceedings of the transportation research board 99th annual meeting*. Transportation Research Board Washington, DC.

12. **Zhou, H.**, & Laval, J. (2020). Hybrid modeling of mandatory lane changes near freeway diverges. In *Proceedings of the transportation research board 99th annual meeting*. Transportation Research Board Washington, DC.

13. **Zhou, H.**, Wang, J., & Cheng, J. (2019). Comparative study of the operating characteristics between cars and trucks on freeway loops: An empirical study based on drone videos. In *Proceedings of the transportation research board 98th annual meeting*. Transportation Research Board Washington, DC.

14. Wang, S., Chen, C., & **Zhou, H.** (2018). Simulation-based research on the comprehensive indexes for highway hypnosis evaluation. In *Cictp 2018: Intelligence, connectivity, and mobility* (pp. 2076–2086). American Society of Civil Engineers Reston, VA.

15. **Zhou, H.**, Wang, Q., Jing, W., Guo, L., & Cheng, J. (2018). Development of the database for domestic system interchanges. In *Cictp 2017: Transportation reform and change—equity, inclusiveness, sharing, and innovation* (pp. 280–290). American Society of Civil Engineers Reston, VA.

16. **Zhou, H.**, Ma, Y., & Cheng, J. (2016). The minimum highway horizontal radius considering the bernoulli effect on running vehicles. In *Cictp 2016* (pp. 1740–1751).

Professional Experience

Editorial positions

2024-2026 — **Editorial board member** of the Journal of Urban Lifeline published by Springer

Committee positions

2023-2026 — **Co-chair** of Technical Committee JT04-Traffic Control of Traffic Engineering Division of the 3rd WTC Academic Sub-Committee

Youth Member of Technical Committee JC0501-CAV & Highway Traffic of Cross-Cutting Division of the 3rd WTC Academic Sub-Committee

Project experience

2020-2022 — **Major student contributor** CPS #1932451: Emulating Emerging Autonomous Vehicle Technologies to Understand Their Impact on Urban Congestion (PI: Jorge Laval, Co-PI: Srinivas Peeta)

2020-2021 — **Major student contributor** NSF# 1826162: Understanding the Impacts of Automated Vehicles on Traffic Flow Using Empirical Data (PI: Danjue Chen, Co-PI: Jorge Laval)

Professional service

2021-2022 — **Webinar coordinator** of TRB-ACP50 (standing committee for traffic flow theory) webinar series on Adaptive Cruise Control/Connected and Automated Vehicles. `https://web.engr.oregonstate.edu/~bertinir/TFT`

2021- — **Journal referee** for Transportation Research Part B, Part C; Transportmetrica A: Transport Science; IEEE-ITS; Transportation Research Record.

Invited talks

Oct 2023 — **Informs 2023:** Fractal-like growth of urban congestion and its control based on deep reinforcement learning.

Aug 2023 — **FAI Summit Panel Talk:** Significance of low-level control on string stability under adaptive cruise control.

Jan 2022 — **TRB podium presentation:** A Novel Experimental Method to Collected Detailed Data from Level-2 Self-driving Cars.

Professional Experience (continued)

Dec 2021 — **TFTC ACC Webinar:** ACCs Are Not All Blackboxes: Insights of Open-sourced Code in Openpilot, Comma.ai.

Patent application

2022 — **Two U.S. Patents (pending) of self-driving technologies** , Those two patents are informed by celebrated car following models but compatible with existing level-2 self-driving cars. They have been filed to the U.S. patent office by Georgia Tech.

Awards

2023 — **Sigma Xi Research Awards Nominee**, CEE School's nominee for Sigma Xi Best Ph.D. Thesis award (pending the final award) at Georgia Tech Chapter.

2020 — **ASHE Babs Abubakari Scholarship**, 2019/2020 Babs Abubakari Scholarship in recognition of student's academic excellence and leadership, awarded by ASHE (American Society of Highway Engineering) at Georgia Chapter.

2015 — **Southeast University President's Award**, for outstanding undergraduates (Top 1% University-wide)

Teaching and leadership

2024 — **Instructor,** Traffic flow theory

2023 — **Instructor,** CUTR Transportation Seminars

2022 — **TA**, giving 6 lectures for the graduate course "Traffic Flow Theory" (CEE 6636) with a focus on the model and impacts of autonomous vehicles.

2019-2020 — **TA**, teaching machine Learning application in Transportation for the graduate course "Simulations in Transport" (CEE 6632); create animation materials used for teaching undergraduate statistics course (CEE 3770).

2019 — **President** of the ASHE (American Society of Highway Engineering) Student Chapter at Georgia Tech

2015-2017 — **Instructor**, build the curriculum, prepare the test filed, and instruct undergraduate students in 2-week road survey field practice affiliated with the undergraduate course "Road Survey and Design"

Skills

- **Self-driving car experiments:** Rich experience in data collection from market cars with GPS/CAN Bus logger; efficient in running customized control algorithms on market self-driving cars with advanced driver assistance systems; DIY development of self-driving control algorithmsl; open-sourced several branches of self-driving longitudinal motion planners based on Comma.AI @ `https://github.com/HaoZhouGT`
- **Traffic and complex system simulation:** Efficient in simulating congested urban networks and other complex systems with fractal geometries in Civil Engineering.

References

Prof. Jorge Laval
Professor (Advisor), Georgia Tech
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Prof. Patricia Mokhtarian
Professor, Georgia Tech
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Prof. Srinivas Peeta
Professor, Georgia Tech
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